# Mid-Term Examination

**Course:** Tools & Techniques of Bioinformatics
**Duration:** 1.00 Hours
**Total Marks:** 20

## Section – A (Answer all the questions) (4*3 = 12)

1. Compare these two primers and determine which one is superior, providing justification for your choice.

**Primer pair 1**

| | Sequence (5'->3') | Template strand | Length | Start | Stop | Tm | GC% | Self complementarity | Self 3' complementarity |
|---|---|---|---|---|---|---|---|---|---|
| Forward primer | GCTAACGTATCCACGCCGTA | Plus | 20 | 1694540 | 1694559 | 59.97 | 55.00 | 5.00 | 3.00 |
| Reverse primer | CATGAAGATGCGGACTTGCG | Minus | 20 | 1695186 | 1695167 | 59.97 | 55.00 | 4.00 | 2.00 |
| Product length | 647 | | | | | | | | |

**Primer pair 4**

| | Sequence (5'->3') | Template strand | Length | Start | Stop | Tm | GC% | Self complementarity | Self 3' complementarity |
|---|---|---|---|---|---|---|---|---|---|
| Forward primer | ATAACGGTTCAGGCACAGCA | Plus | 20 | 1694668 | 1694687 | 59.96 | 50.00 | 5.00 | 0.00 |
| Reverse primer | AGGTGGTTGCAACTGGACAA | Minus | 20 | 1695407 | 1695388 | 60.03 | 50.00 | 8.00 | 3.00 |
| Product length | 740 | | | | | | | | |

2. Compare the three PDB structures (PDB IDs: 4DPZ, 8BWG, and 8CNJ) and determine which one is the most suitable. Provide a rationale for your selection.
3. Compare the ADME (Absorption, Distribution, Metabolism, and Excretion) and toxicity properties of the three ligands (CID311, CID6508, and CID243). Based on your analysis, provide your opinion on which ligand is the most suitable drug candidate.
4. Generate a 2D diagram of the specified protein-ligand interaction and describe the interaction, including the number of bonds involved. (Link: https://drive.google.com/file/d/1COWT6-xnsChU9iagUrcXHfQqOOx_Xccd/view?usp=sharing )

## Section – B (Answer any four questions) (4*2 = 8)

1. How does bioinformatics integrate multiple disciplines like computer science, biology, and statistics?
2. How does the compound selection process in Discovery Research impact the efficiency of drug development, and what role do computational methods play in narrowing down compounds?
3. What is the significance of molecular dynamics simulations in evaluating the interaction between a drug and its target protein?
4. How do RMSD and RMSF analyses together inform decisions on ligand modifications to enhance stability?
5. What role does Radius of Gyration analysis play in evaluating the compactness and stability of a protein-ligand complex during simulations?